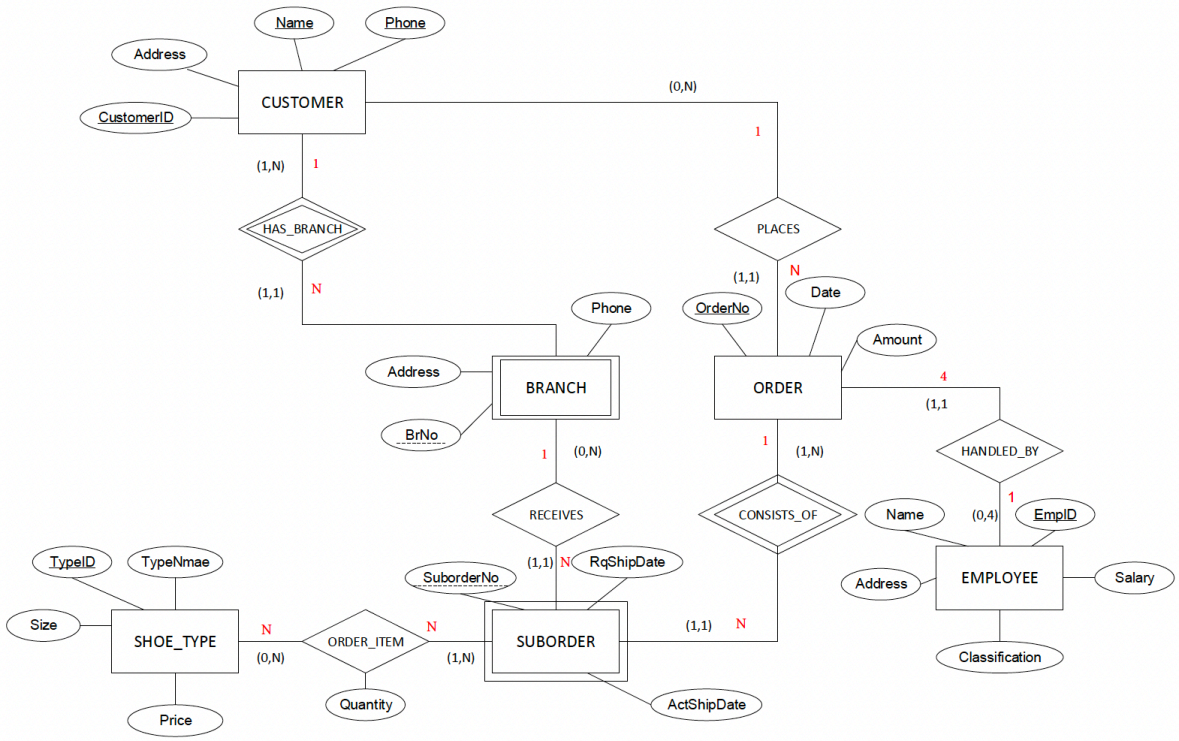
Lab 2

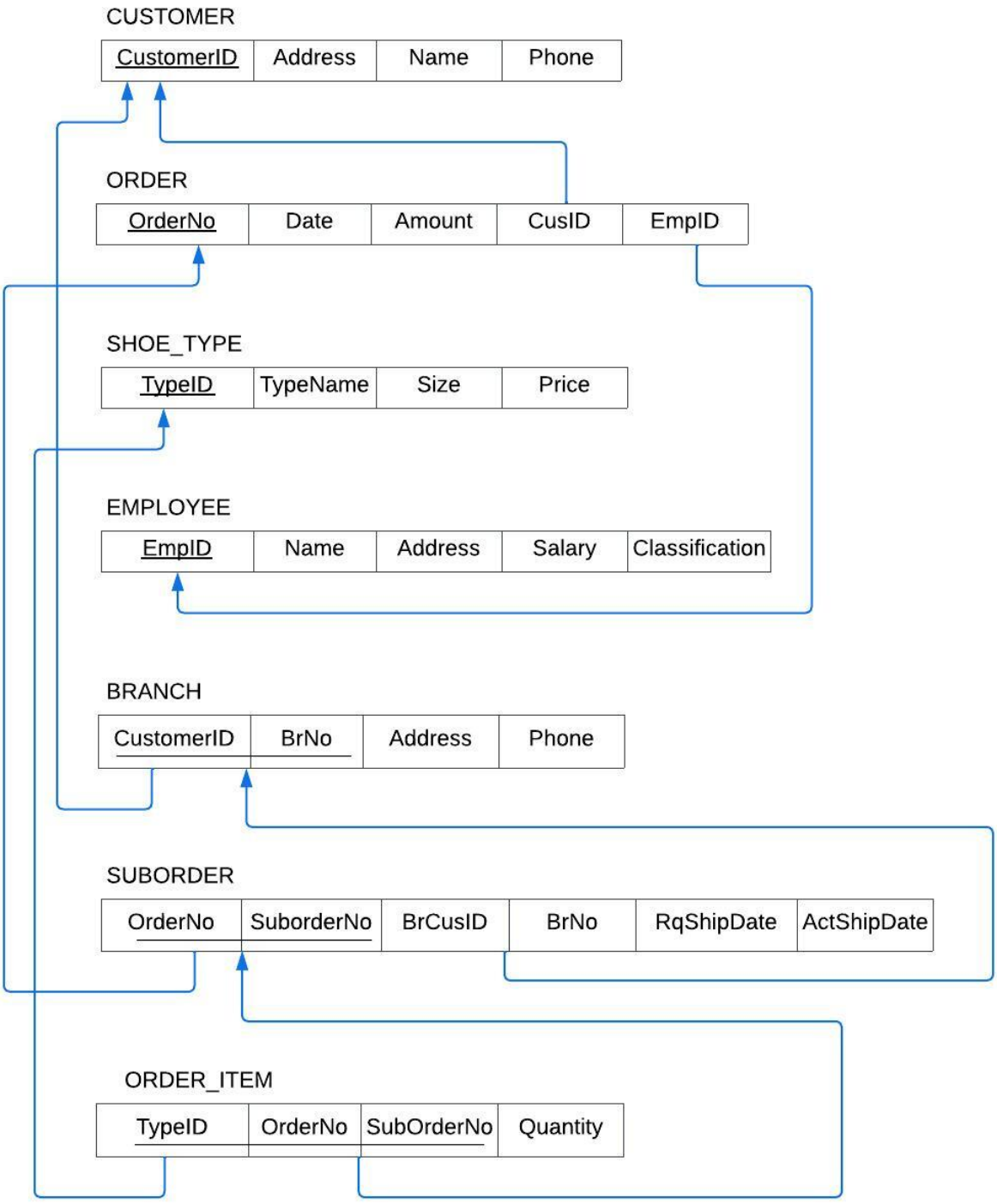
An Assignment for class **MET CS 579** Database Management at Boston University due on February 23, 2023, instructed by Professor **Jae Lee**. Students Wei Wang and Jin Yu Zhang submitted on February 22, 2023.

Constraints

Constraints which were specified in the ER model but not reflected in the relational database schema:

1. In the ER model, the cardinality ratio between ORDER and EMPLOYEE is 4:1. However, in the relational database schema it could only be reflected as N:1.
2. There are 3 candidate keys (CustomerID, Name, Phone) in CUSTOMER entity, and we arbitrarily selected CustomerID as the primary key.
3. In Customer relation, Name and Phone are keys, so they should be set as Unique and Not null.
4. In Order relation, EmpID and CusID should be set as Not null, because the minimum number in Min-Max notation is 1.
5. In Suborder relation, BrCusID and BrNo should be set as Not null, because the minimum number in Min-Max notation is 1.


Name
Phone
Address
CUSTOMER
CustomerID
(0,N)
1
(1,N)
1
HAS_BRANCH
PLACES
(1,1)
N
(1,1)
N
Phone
OrderNo
Date
Amount
Address
BRANCH
ORDER
4
(1,1
BrNo
1
(0,N)
1
(1,N)
HANDLED_BY
RECEIVES
CONSISTS_OF
Name
(0,4)
EmplID
1
TypeID
TypeNmae
(1,1)
N
SuborderNo
RqShipDate
Address
EMPLOYEE
Salary
Size
SHOE_TYPE
N
ORDER_ITEM
N
SUBORDER
(1,1)
N
(0,N)
(1,N)
Classification
Quantity
Price
ActShipDate


CUSTOMER
CustomerID
Address
Name
Phone
ORDER
OrderNo
Date
Amount
CusID
EmpID
SHOE_TYPE
TypeID
TypeName
Size
Price
EMPLOYEE
EmpID
Name
Address
Salary
Classification
BRANCH
CustomerID
BrNo
Address
Phone
SUBORDER
OrderNo
SuborderNo
BrCusID
BrNo
RqShipDate
ActShipDate
ORDER_ITEM
TypeID
OrderNo
SubOrderNo
Quantity